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MTECH [AISP] Subject : DIVP
TA: 3

Q.1) Consider the images shown. The image on the right was obtained by :
a) multiplying the image on the left by $(-1)^{x+y}$
b) computing the DFT
c) taking the complex conjugate of the transform.
d) computing the inverse DFT
e) multiplying the real part of the result by $(-1)^{x+y}$
Explain (mathematically) why the image on the right appears as it does.

→ Let the original image be $f(x,y)$ of size $M \times N$
Let its 2-DFT be $F(u,v)$
Where

$$F(u,v) = \sum_{x=0}^{M-1} \sum_{y=0}^{N-1} f(x,y) e^{-j2\pi\left(\frac{ux}{M} + \frac{vy}{N}\right)}$$

and
inverse DFT is

$$f(x,y) = \frac{1}{MN} \sum_{u=0}^{M-1} \sum_{v=0}^{N-1} F(u,v) e^{j2\pi\left(\frac{ux}{M} + \frac{vy}{N}\right)}$$

Assume that, M & N are even

a) Multiplication by $(-1)^{x+y}$

$$g(x,y) = f(x,y)(-1)^{x+y}$$

$$(-1)^x = e^{j\pi x}$$

$$(-1)^y = e^{j\pi y}$$

$$(-1)^{x+y} = e^{j\pi(x+y)}$$

$$g(x,y) = f(x,y)\, e^{j\pi x}\, e^{j\pi y}$$

Using the frequency shifting property

$$G_e(u,v) = f\left(u - \frac{M}{2}, v - \frac{N}{2}\right)$$

Thus, multiplying the image by $(-1)^{x+y}$ shifts its fourier spectrum by

$$\left(\frac{M}{2}, \frac{N}{2}\right)$$

b) Computing the DFT

Now compute the DFT of $g(x,y)$

$$G(u,v) = f\{g(x,y)\}$$

From the above,

$$G(u,v) = f\left(u - \frac{M}{2}, v - \frac{N}{2}\right)$$

The spectrum has been shifted by $\left(\frac{M}{2}, \frac{N}{2}\right)$

c) Taking the complex conjugate

$H(u,v) = G_e^*(u,v)$ imp DFT property is

$$f^{-1}\{F^*(u,v)\} = f^*(-x,-y)$$

Where the coordinates are interpreted modulo M & N

Therefore when we take the complex conjugate of the transform and the perform the inverse DFT, the spatial image becomes reflected about both axes since the original image is a real image

$$f^*(x,y) = f(x,y)$$

$f^*(-x,-y) = f(-x,-y)$

So image undergoes a 180° rotation

d) Computing the Inverse DFT, we now calculate

$$h(x,y) = F^{-1}\{G^*(u,v)\}$$

Using the conjugate property

$$h(x,y) = g^*(-x,-y)$$

Because the image is real

$$g^*(-x,-y) = g(-x,-y)$$

Recall that

$$g(x,y) = (-1)^{x+y} f(x,y)$$

Therefore

$$h(x,y) = (-1)^{-x-y} f(x,-y)$$

$$(-1)^{-x-y} = (-1)^{x+y}$$

we get,

$$h(x,y) = (-1)^{x+y} f(-x,-y)$$

e) Multiplying the Real part by $(-1)^{x+y}$

$$r(x,y) = (-1)^{x+y} h(x,y)$$

Substitute the expression for $h(x,y)$

$$r(x,y) = (-1)^{x+y}\left[(-1)^{x+y} f(-x,-y)\right]$$

$$r(x,y) = (-1)^{2x+2y} f(-x-y)$$

$$(-1)^{2x+2y} = 1$$

we obtain

$$r(x,y) = f(-x,-y)$$

* Properties of DFT are used

1. Frequency Shifting / centering Property

$$g(x,y) = f(x,y)(-1)^{x+y}$$

then DFT becomes

$$G(u,v) = F\left(u - \frac{M}{2}, v - \frac{N}{2}\right)$$

2. Complex Conjugation Property

$$f(x,y) \leftrightarrow F(u,v)$$
$$F^*(u,v) \leftrightarrow F^*(-x,-y)$$
$$f^{-1}\{F^*(u,v)\} = f^*(-x,-y)$$

3. Real Image Property

$$f^*(x,y) = f(x,y)$$
$$f^*(-x,-y) = f(-x,y)$$

4. Multiplication Property / Frequency shifting Again

$$(-1)^{x+y}(-1)^{x+y} = (-1)^{2x+2y} = 1$$
$$f(x,y) \rightarrow f(-x,-y)$$

which is 180° rotation

(Q.2) Consider a 3×3 spatial kernel that average the four closest neighbors of a point (x,y) but excludes the point itself from the average.

a) Find the equivalent filter transfer function H(u,v) in frequency domain
b) Show that your result is a lowpass filter transfer function

→ four closest neighbors of pixel (x,y)
Top: (x-1, y)
Bottom: (x+1, y)
Left: (x, y-1)
Right: (x, y+1)

3×3 kernel

$$h(x,y) = \frac{1}{4}\begin{bmatrix} 0 & 1 & 0 \\ 1 & 0 & 1 \\ 0 & 1 & 0 \end{bmatrix}$$

a) for a spatial filter, the frequency response is obtained by taking the fourier transform of kernel

The four non zero term are located at
(-1, 0), (1, 0), (0, -1), (0, 1)
Using the 2D fourier transform

$$H(u,v) = \sum_x \sum_y h(x,y) e^{-2j\pi\left(\frac{ux}{m} + \frac{vy}{N}\right)}$$

Substitute the four non zer values

1. $(x,y) = (-1,0)$

$$h(x,y) = h(-1,0) = \frac{1}{4} e^{-j2\pi\left(\frac{-u}{M}\right)}$$

$$h(-1,0) = \frac{1}{4} e^{j2\pi u/M}$$

2. $(x,y) = (1,0)$

$$h(1,0) = \frac{1}{4} e^{j2\pi u/M}$$

3. $(x,y) = (0,-1)$

$$h(0,-1) = \frac{1}{4} e^{j2\pi v/N}$$

4. $(x,y) = (0,1)$

$$h(0,1) = \frac{1}{4} e^{-j2\pi v/N}$$

$$H(u,v) = \frac{1}{4}\left[e^{j2\pi u/M} + e^{-j2\pi u/M} + e^{j2\pi v/N} + e^{-j2\pi v/N} \right]$$

By Eulers Identity,

$$e^{j\theta} + e^{-j\theta} = 2\cos\theta$$

we got,

$$H(u,v) = \frac{1}{4}\left[2\cos\left(\frac{2\pi u}{M}\right) + 2\cos\left(\frac{2\pi v}{N}\right) \right]$$

$$\therefore H(u,v) = \frac{1}{2}\left[\cos\left(\frac{2\pi u}{M}\right) + \cos\left(\frac{2\pi v}{N}\right) \right]$$

b) i) Response at zero frequency

$u = 0, v = 0$

$$H(0,0) = \frac{1}{2}[\cos(0) + \cos(0)]$$

$\cos(0) = 1$, $H(0,0) = \frac{1}{2}(1+1) = 1$

$\therefore H(0,0) = 1$

This means the DC component & very low frequency components are passed strongly

2) Behaviour at low frequencies

$$\cos\theta = 1 - \frac{\theta^2}{2}$$

$$\cos\left(\frac{2\pi u}{M}\right) \approx 1$$

$$\cos\left(\frac{2\pi v}{N}\right) \approx 1$$

$$H(u,v) \approx 1$$

near the origin

Therefore low frequency components are passed with high gain

3) Effect on Rapid Image variations

High Spatial frequencies corresponds to rapid changes in intensity such as

- sharp edges
- Fine details
- alternating pixel pattern

The averaging operation reduces many such rapid variations because each pixel is replaced by an average of its neighbors. Thus, the filter peforms a smoothing operation

Smoothing is associated with low pass filtering. Therefore,
Four-neighbor averaging => smoothing => low pass frequency emphasis.